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Synchronization between two linearly coupled reservoir computers

Wancheng Hu, Yibin Zhang, Rencai Ma, Qionglin Dai*, Junzhong Yang

School of Science, Beijing University of Posts and Telecommunications, Beijing 100876, China

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ABSTRACT

Reservoir computing, using a model-free algorithm, has drawn considerable attention in the past few years. In this work, we use a reservoir computer to reconstruct the attractor of its learned chaotic system. We present a form of interaction between two reservoir computers inspired by the diffusion coupling of oscillators. By mixing the readout of two reservoir computers with an appropriate coupling strength, we present the complete synchronization of reservoir computers. Numerical experiments are performed to verify the theory.

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1. Introduction

Reservoir computer [1–4], one type of recurrent neural networks in machine learning, has shown favourable performance in various tasks in past two decades such as predicting the time series of chaotic systems [5–7] and inferring unmeasured variables by using measured variables of a dynamical system [8,9]. A reservoir computer contains an input layer, an output layer and a reservoir consisting of a sparsely connected networks. Different from recurrent neural networks, only the output layer has to be trained while the input layer and reservoir stay unchanged during the learning. After the output layer has been trained, a reservoir computer can process the input time series to fit the task. Reservoir computing is a model-free method, that is, when we want to diagnose the ergodic properties of the underlying processes producing the data, we can do so based only on the data itself [7]. Furthermore, it has been thought that reservoir computers perform better when the dynamics of reservoir are at the edge of criticality [10,11], which allows a physical implement of reservoir computers. The physical reservoirs include a time-delay electronic circuit [12], optical networks of semiconductor lasers [13], an electro-optic system [14,15] and a field-programmable gate array [16].

Recently, reservoir computing has been applied to the investigation in the field of nonlinear dynamics [17]. Prediction on a system's behavior without its full knowledge is always a crucial scientific problem [18]. Starting from the same initial conditions, reservoir computers may generate dynamical trajectory staying closed to the true chaotic one in several Lyapunov time after being trained [7]. Zhang *et al.* showed that reservoir computer can be exploited for long-term prediction of the phase of chaotic oscillators due to the phase information is related to the zero Lyapunov exponent [19]. Due to its great ability to process time series signals, a reservoir computer may be trained to learn from a chaotic system. Pathak *et al.* have trained reservoir computer to approximate the ergodic properties of the true systems such as those with positive or zero Lyapunov exponents [7]. Lu *et al.* proved that reservoir computers may reconstruct the attractors for chaotic processes from limit time series data and ascribed it to the mechanism of generalized synchronization [20].

Synchronization among interacting oscillators is one of the most important collective behaviors and is well recognized in variety of systems from physical, chemical, to biological and human systems [21–24]. Oscillators are in synchronization when they behave at a same rhythm. Different types of synchronization in coupled chaotic oscillators have been recognized, for example, complete synchronization [25,26], phase synchronization [27,28], and generalized synchronization [29,30]. A reservoir computer may be treated as an autonomous dynamical system when its output is used to be input signal. What happens to synchronization when reservoir computers are involved? Weng *et al.* realized the synchronization between a well-trained reservoir computer and its learned chaotic system by linking them with a common signal [31]. Now one question arises: can synchronization be realized when reservoir computers trained by the chaotic oscillators are coupled together?

The rest of paper is organized as follows. In Section 2, we show the structure of a reservoir computer and how reservoir computers are trained. The coupling between two computers are introduced. In Section 3, we train two sets of computers by Rössler and Lorenz

* Corresponding author.
E-mail addresses: qldai@bupt.edu.cn (Q. Dai), jzyang@bupt.edu.cn (J. Yang).

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systems, respectively. The synchronization between two computers are investigated. Finally, we make a conclusion in Section 4.

2. Model

As shown in Fig. 1, the basic structure of the reservoir computer contains an input layer, a linear output layer, and a reservoir network. The reservoir network is a large, low-degree (d), directed Erdös-Rényi network with a N×N adjacency matrix A, appropriately scaled so that its largest eigenvalue is equal to ρ. It has the same topology features with a general Erdös-Rényi network. The corresponding network parameters are listed in Table 1. When a reservoir computer is trained, an input signal $\boldsymbol{u}(t)$ is continuously input into the reservoir network with an input weight matrix $\boldsymbol{W}_{in}$ which is a random matrix contains elements uniformly distributed in $[-1, 1]$. Then the state vector $\boldsymbol{r}$ of the reservoir network is updated according to following equation:

$$\boldsymbol{r}(t+1) = \tanh[\boldsymbol{A}\boldsymbol{r}(t) + \boldsymbol{W}_{in}\boldsymbol{u}(t)], \tag{1}$$

where $\boldsymbol{A}$ is the connection matrix of the reservoir network [5]. The non-zero elements of matrix $\boldsymbol{A}$ are randomly selected from the uniform distribution $[-a, a]$. The value of a can be changed to get an appropriate eigenvalue, also called spectral radius, $\boldsymbol{\rho}$ of the matrix $\boldsymbol{A}$. With the input signal $\boldsymbol{u}(t)$ and the state vector $\boldsymbol{r}(t)$, the output of the reservoir computer takes the form

$$\boldsymbol{y}(t) = \boldsymbol{W}_{out}\begin{bmatrix}\boldsymbol{u}(t)\\ \boldsymbol{r}(t)\end{bmatrix}, \tag{2}$$

where $\boldsymbol{W}_{out}$ is the output weight matrix.

During the training period, an input signal $\boldsymbol{u}(t)$ is fed into the reservoir and the goal is to approximate the desired input-related time series by training $\boldsymbol{W}_{out}$. To do it, $\boldsymbol{r}$(t) and $\boldsymbol{u}$(t) are recorded into two matrices $\boldsymbol{X}$ and $\boldsymbol{Y}$, which take the forms of [$\boldsymbol{u}$(k); $\boldsymbol{r}$(k)] and [$\boldsymbol{u}$(k+1)] in their $\boldsymbol{k}_{\text{th}}$ column, respectively. Then $\boldsymbol{W}_{out}$ is calculated according to the ridge regression, which is given by

$$\boldsymbol{W}_{out} = \boldsymbol{Y}\boldsymbol{X}^T\left(\boldsymbol{X}\boldsymbol{X}^T + \lambda\mathbb{I}\right)^{-1}, \tag{3}$$

Table 1
Reservoir parameters for the simulation of chaotic systems.

For Rössler system	For Lorenz system
N=500	N=800
d=20	d=20
T=100	T=100
Δt=0.1	Δt=0.02
ρ=1.8	ρ=1.2
a=0.1	a=0.1

where $\mathbb{I}$ is an identity matrix and λ is the ridge regression parameter to avoid over-fitting. After training, the input $\boldsymbol{u}(t)$ is replaced by the output $\boldsymbol{y}(t)$, that is, $\boldsymbol{y}(t)$ is fed back into the reservoir as the input signal. Then a trained reservoir becomes an autonomous dynamical system which mimics the system from which $\boldsymbol{u}(t)$ comes. Following this way, we train two reservoir computers.

To consider the synchronization between reservoir computers, we need to establish the interaction between them. Inspired by the diffusion coupling of oscillators, we consider the input to each reservoir computer taking the form

$$\boldsymbol{u}_{1,2}(t+1) = \alpha\boldsymbol{y}_{2,1}(t) + (1-\alpha)\boldsymbol{y}_{1,2}(t), \tag{4}$$

where $\boldsymbol{y}_{1,2}$ represent the output of the reservoir computers 1 and 2.

Through this kind of input, the mixture of their outputs, each reservoir computer is partly driven by the other and the interaction between them is built. When $\alpha = 0$, the two reservoirs evolve independently and, when $\alpha = 1$, the coupling between the two computers becomes one-way coupling. To be stressed, the parameter α characterizes the coupling method between two computers and measures the coupling strength between the two computers.

3. Numerical results

3.1. Attractors reconstruction

We consider the synchronization between reservoir computers trained by Rössler system or Lorenz system. The Rössler system is given by

$$\begin{aligned}\frac{dx}{dt} &= -y - z,\\ \frac{dy}{dt} &= x + ay,\\ \frac{dz}{dt} &= b + z(x - c),\end{aligned} \tag{5}$$

with the parameters $a = 0.5$, $b = 2$, and $c = 4$. The Lorenz system is given by

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x),\\ \frac{dy}{dt} &= x(r - z) - y,\\ \frac{dz}{dt} &= xy - bz,\end{aligned} \tag{6}$$

with $\sigma = 10$, $r = 28$, and $b = 1$. Under these parameters, both Rössler system and Lorenz system allow for chaotic dynamics.

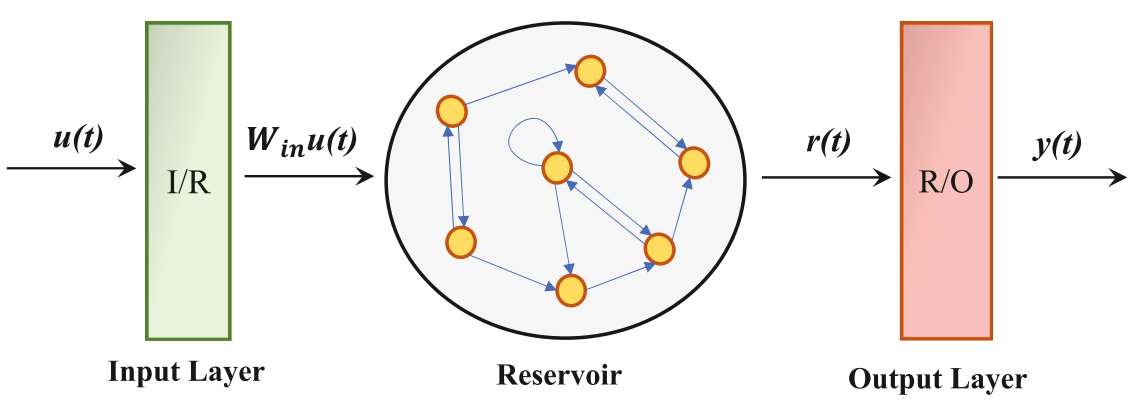

Fig. 1. A reservoir computer includes three components: an input layer, a linear output layer and a reservoir network. The reservoir network consists of N randomly sparsely connected nodes.

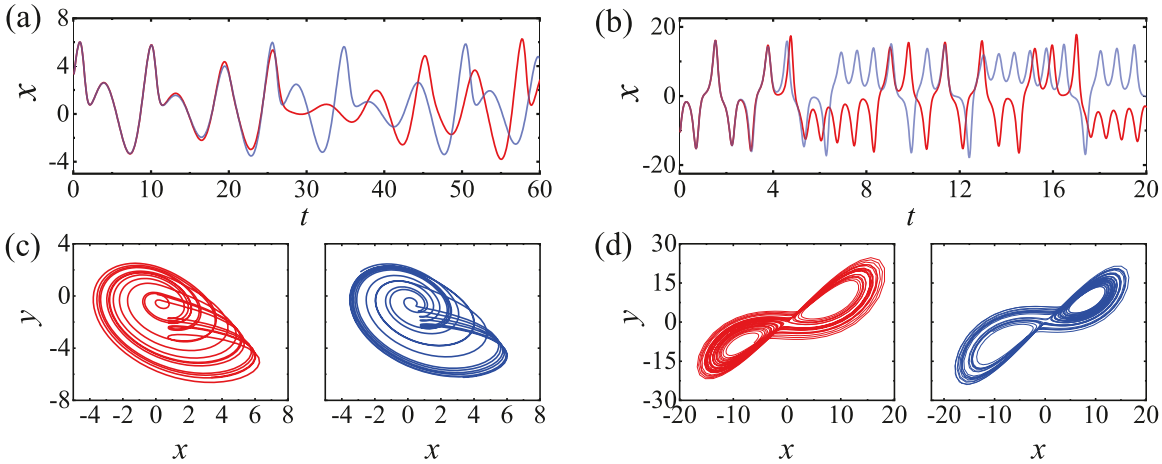

Fig. 2. The time sequences and the phase diagrams in the $x-y$ plane for chaotic systems and the trained reservoir computers. (a,b) Predicted (blue) and actual (red) $x(t)$ for chaotic Rössler and Lorenz systems, respectively, which reflect the short-term predicting ability of reservoir computers. (c,d) The phase trajectories in the $x-y$ plane of actual systems (red) and reservoir computers (blue) for Rössler and Lorenz systems, respectively. The similar trajectories indicate that the reservoir computers may simulate the long-term dynamic characteristics of actual chaotic systems.

To train reservoir computers, where the output $\boldsymbol{y}(t)$ contains three elements denoted as $(x(t), y(t), z(t))$, the Rössler system (5) and the Lorenz system (6) are numerically simulated with the 4-th Runge-Kutta method with a time step $\delta t = 0.01$. The input signals $\boldsymbol{u}(t)$ are sampled every $t = 0.1$ time unit from $(x(t), y(t), z(t))$. The total training period is around $T = 100$ time units (The corresponding parameters of reservoir computers are presented in Table I.). After training, the output is reinjected into the computers as input and autonomous reservoir computers are built. With the almost same initial conditions, the reservoir computers may follow target Rössler system and Lorenz system in the first few cycles, which is inversely related to the largest Lyapunov exponent of the target system, and then their time series become uncorrelated, which are presented in Figs. 2 (a) and (b). Since only a few cycles, measured by the inverse of the largest Lyapunov exponent of the target system, can be predicted, we call this situation short-term prediction. Though reservoir computers only possesses short-term prediction ability on the evolution of a chaotic system under given initial conditions, they do reconstruct the asymptotic properties of target systems such as attractor. Figs. 2 (c) and (d) show the results for the Rössler systems and the Lorentz systems, respectively. Clearly, the phase trajectories in the $x-y$ plane, generated by reservoir computers and target systems, show no difference between them. The asymptotic dynamical behaviors from reservoir computers and target systems can be further studied by the power spectrum of the time series and the Poincáre return maps, and the same conclusion can be drawn [20]. In short, once trained, reservoir computers may be treated as autonomous dynamical systems.

3.2. Coupled reservoir computers can achieve synchronization

As illustrated in Section 3.1, trained reservoir computers can act as autonomous dynamical systems which display similar asymptotic behaviors to their target systems. It is well known that coupled dynamical systems may establish the synchronization between them under proper conditions. Now, we are interested in investigating the synchronization between trained reservoir computers. We consider the complete synchronization [26] where $\boldsymbol{y}_1(t) \approx \boldsymbol{y}_2(t)$ is held when $t \to \infty$ between two reservoir computers which are trained by the same dynamical systems. We suppose that the two reservoir computers are coupled with each other according to Eq. (4).

We first consider autonomous reservoir computers trained by the Rössler system (5). When the coupling strength α is sufficiently low, for example $\alpha = 0.005$, the two reservoir computers are not in complete synchronization. As shown by the their time series in Fig. 3(a), the two computers are desynchronized both in their amplitudes and phases. On the other hand, their phase trajectories in the $x-y$ plane indicate that the two computers are on the same attractor. Then, we increase α to $\alpha = 0.0057$. The two computers are not in complete synchronization. However, the time series in Fig. 3(b) suggest that the phases of the two computers are in phase but their amplitudes display the characteristics of the on-off intermittency. For $\alpha = 0.5$, we find the complete synchronization between two computers in Fig. 3(c) both in phase and in amplitude. Further increasing α to $\alpha = 0.83$, the coupling becomes close to one-way coupling and the mutual coupling between the two computers is weakened. Consequently, the complete synchronization between the two computers is lost. Actually, Fig. 3(d) shows that the synchronization evolves to a state of equilibrium. Similar results can be found for the coupled reservoir computers trained by the Lorenz system (6). As shown in Fig. 4, the complete synchronization can be reached at intermediate α while desynchronization at α close to zero. When α is close to 1, the two reservoir computers jump to the state of equilibrium.

To get an overview on the synchronization between two reservoir computers, we measure the synchronization between them. For this aim, we consider the synchronization error, which is defined as

$$\begin{aligned}\Delta &= \langle |\boldsymbol{y}_1(t) - \boldsymbol{y}_2(t)| \rangle \\ &= \langle \sqrt{[x_1(t)-x_2(t)]^2 + [y_1(t)-y_2(t)]^2 + [z_1(t)-z_2(t)]^2} \rangle,\end{aligned} \tag{7}$$

where $\langle \cdot \rangle$ means the average over time. When the synchronization between two reservoir computers is built, the synchronization error Δ reaches a very small value. As shown in Fig. 5, the synchronization error Δ stays at low values in the most part of intermediate α, which suggests the synchronization between the reservoir computers. Actually, there are couple of transitions to synchronization. Firstly, with the increase of α from zero, the synchronization error Δ drops from a high value, i.e., $\Delta > 1$ to a lower one, $\Delta < 0.01$, which suggests the onset of synchronization. However, compared with Figs. 3 and 4, the synchronization error cannot distinguish the complete synchronization from the phase synchronization. Secondly, at high α, there exists a sharp desynchro-

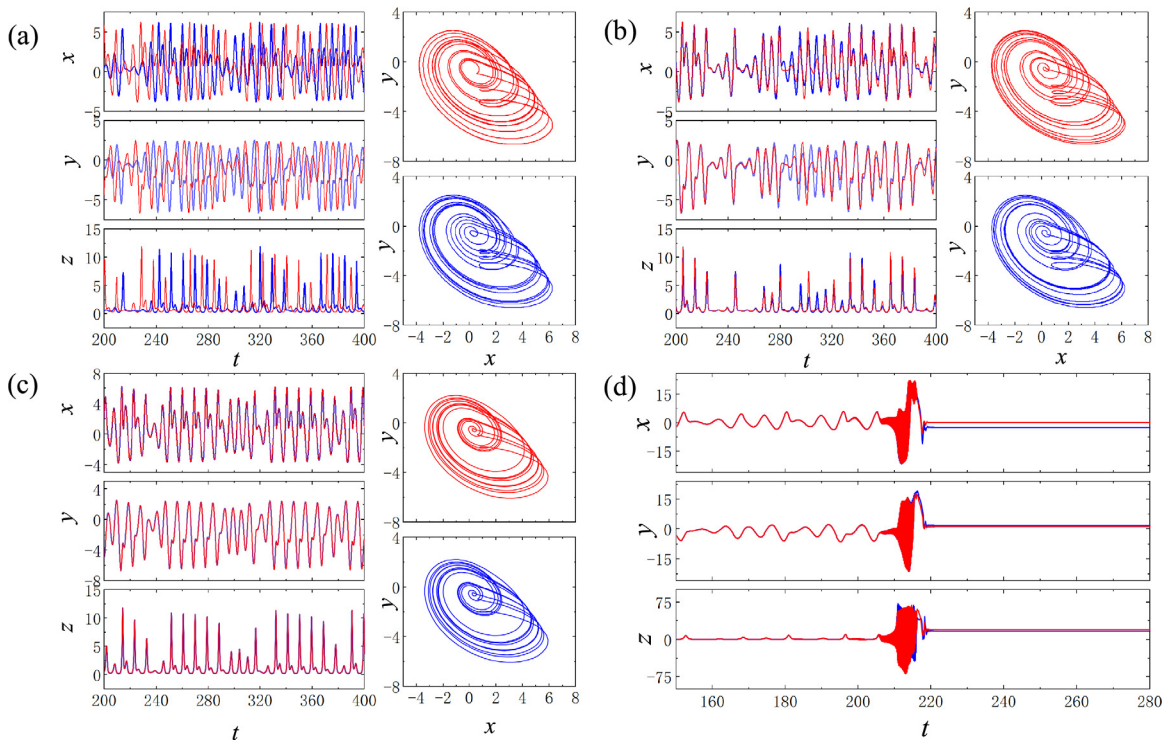

Fig. 3. The synchronization between two reservoir computers trained by the same chaotic Rössler systems. (a) $\alpha = 0.005$, (b) $\alpha = 0.0057$, and (c) $\alpha = 0.5$. In each subplot, the panels in the left column show the time series of $x(t)$, $y(t)$ and $z(t)$ for the two reservoir computers and those in the right column present their phase trajectories in the plane of $x - y$. (d) $\alpha = 0.83$ where the two reservoir computers transition from chaotic dynamics to equilibria.

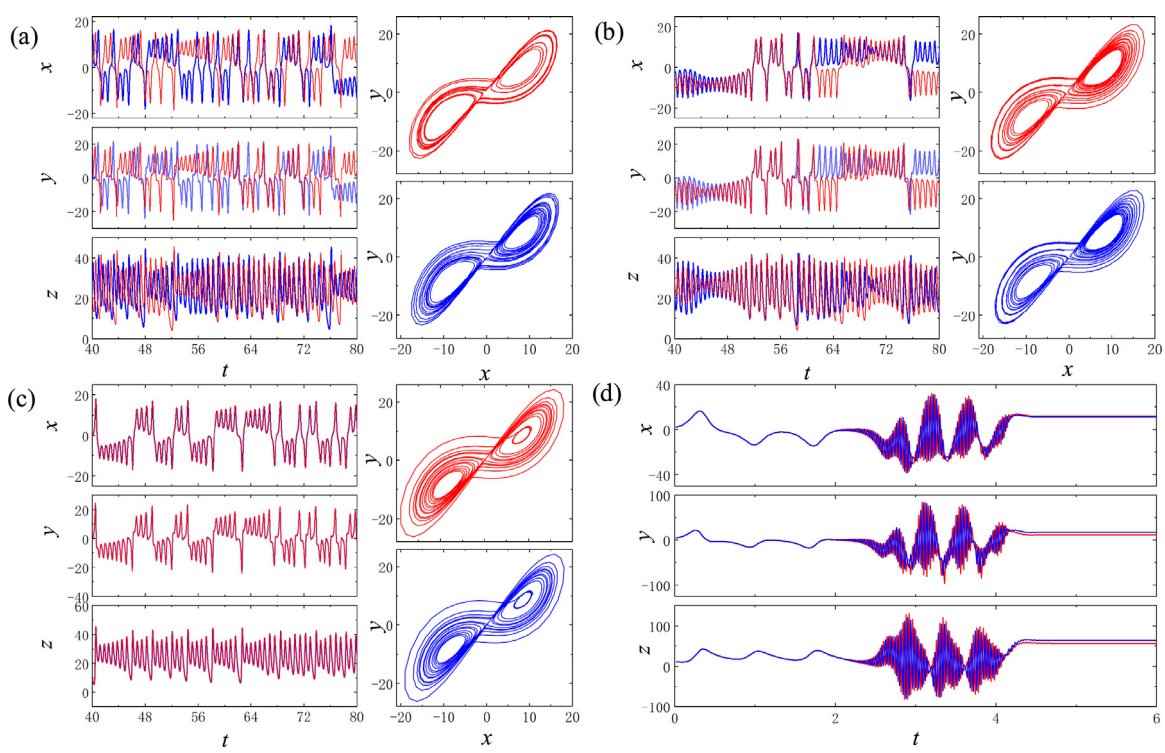

Fig. 4. The synchronization between two reservoir computers trained by the same chaotic Lorenz systems. (a) $\alpha = 0.005$, (b) $\alpha = 0.0085$, and (c) $\alpha = 0.5$. In each subplot, the panels in the left column show the time series of $x(t)$, $y(t)$ and $z(t)$ for the two reservoir computers and those in the right column present their phase trajectories in the plane of $x - y$. (d) $\alpha = 0.85$ where the two reservoir computers transition from chaotic dynamics to equilibria.

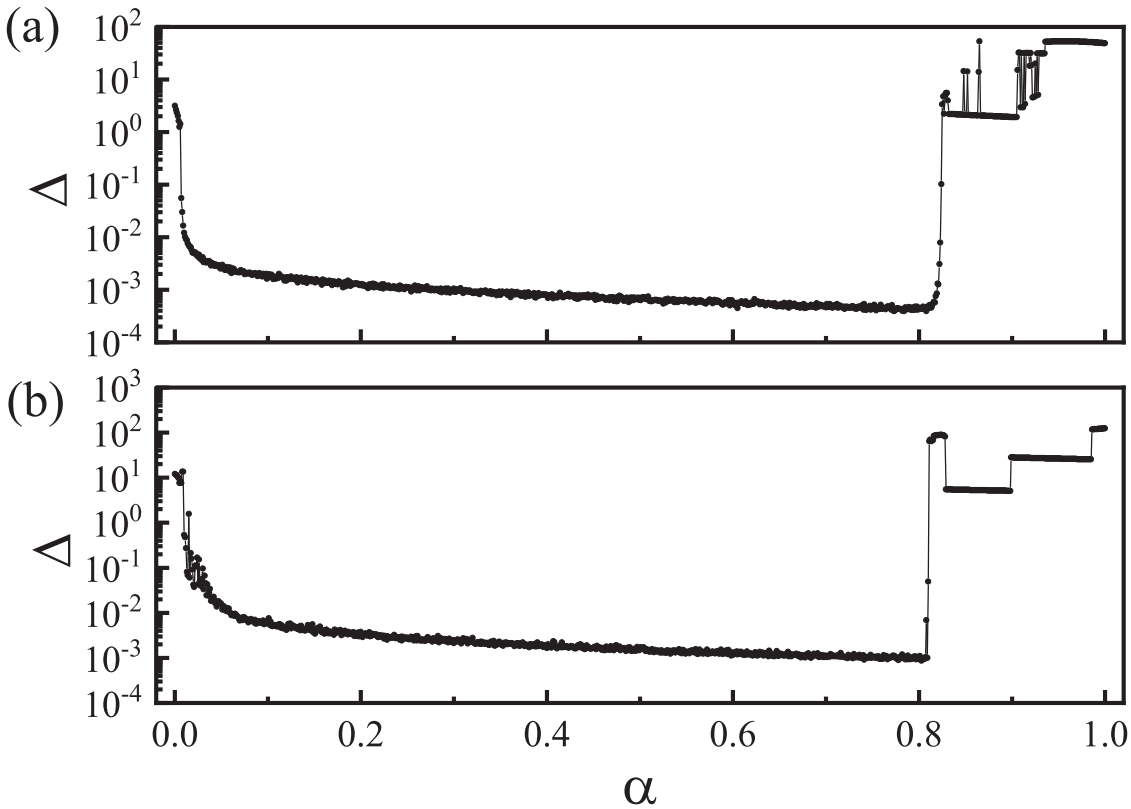

Fig. 5. The synchronization error Δ is plotted against the coupling strength α for Rössler system in (a) and for Lorenz system in (b).

nization transition. At around $\alpha = 0.8$, the synchronization error Δ suddenly jumps to a high value from $\Delta \approx 10^{-3}$, which is induced by the transition of the asymptotic dynamical behavior from chaotic behavior to equilibrium.

4. Conclusion

Here we have studied the synchronization between two reservoir computers. Using Rössler system and Lorenz system, we train reservoir computers, respectively. For a pair of reservoir computers trained by the same dynamical system, we introduce the coupling between them by injecting the mixture of their output. We find that, as the coupling parameter increases, the two reservoir computers may transition from desynchronous to synchronous state and to a state of equilibrium. In this work, we just studied the complete synchronization between reservoir computers trained by the same dynamical systems. However, it can be expected that other types of synchronization such as phase synchronization, generalized synchronization and others can be realized between reservoir computers trained by different dynamical systems in future studies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wancheng Hu: Conceptualization, Methodology, Software, Investigation, Resources, Writing – original draft, Visualization. **Yibin Zhang:** Software, Validation, Investigation, Data curation. **Rencai Ma:** Software, Validation, Investigation, Visualization. **Qionglin Dai:** Investigation, Data curation, Resources, Writing – review & editing, Visualization. **Junzhong Yang:** Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Supervision.

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